02246 Mandatory Assignment
Assignment 03 - Probabilistic Temporal Logics*

To be submitted on DTU Learn - see deadline on DTU Learn

You are encouraged to work in groups, but you must clearly identify the contributions of each group member, and you will be jointly responsible for the finished report. Register your group on DTU Learn before submitting as group submission.

Answers to all parts should be typed up using LaTeX and submitted electronically as a PDF report using the provided template. Drawings and formulae may be handwritten and scanned. More detailed instructions as to the style of answer we expect for each part are included below.

Some tasks require to upload files.

*Thanks to Michael Smith (the original author), and Lijun Zhang, Kebin Zeng, Flemming Nielson, Alberto Lluch Lafuente and Andrea Vandin (contributors).

A03 - Probabilistic Temporal Logics

A03P: Practical Problems

A03P.1 - Loke S. G. Jørgensen In this problem, we will add probabilities to the FCFS scheduler from the previous assignment (the first version provided to you, without the extensions done in previous assignments), so that we construct a discrete time Markov chain.

a) Identify all the sources of non-determinism in the model, and explain whether they are due to local non-determinism between the commands in a module, or due to the concurrent execution of two or more modules.

When a new job is created by either client1 or client2, the job length is chosen nondeterministically. This is local nondeterminism, as there are several transitions with the same label (`create`) and condition (`state=0`). The choice between creating a job for client1 or client2 is also nondeterministic, due to concurrent execution, as both transitions have the same condition (`job2=0`. Finally, if there is a job at the head of the queue while the end of the queue is empty, there is a nondeterministic choice between serving or finishing the job at the head of the queue, and creating a new job, again due to concurrent execution.

b) Resolve the local non-deterministic choices by modifying the modules to include probabilistic commands. Recall that the syntax of this, in PRISM, is as follows:

$$[\langle ACTION \rangle] \ \langle GUARD \rangle \ \rightarrow \ p_1 : \langle UPDATE_1 \rangle + \cdots + p_n : \langle UPDATE_n \rangle \ ;$$

where $\sum_{i=1}^{n} p_i = 1$. For now, you should use a *uniform distribution* (i.e. one where all the probabilities are the same) for each probabilistic command.

```
[create1] state1=0 ->
    1/5 : (state1'=1) & (task1'=1)
  + 1/5 : (state1'=1) & (task1'=2)
  + 1/5 : (state1'=1) & (task1'=3)
  + 1/5 : (state1'=1) & (task1'=4)
  + 1/5 : (state1'=1) & (task1'=5);
```

c) Change the first line of the PRISM file from '`mdp`' to '`dtmc`', and save the model in a new file. This tells PRISM that the model describes a DTMC. Build the model in PRISM, and check that there are no error messages. How many states does your model have? Provide a screenshot.

UPLOAD REQUIRED: the new prism model `A03P1.c.prism`.

The new model has 83 states. See Figure 1.

d) The model you have constructed is purely probabilistic. What has happened to the non-determinism that we had due to the concurrent execution of the modules?

By changing the model type from mdp to dtmc, the non-determinism arising from the concurrent execution of modules is resolved probabilistically. Whenever multiple commands are enabled, PRISM selects between them using equal probabilities. Consequently, choices such as whether serve1 or create2 occurs first are no longer nondeterministic; they become probabilistic transitions in the DTMC. The probabilistic choices of job length are also retained, so the resulting model is purely probabilistic.

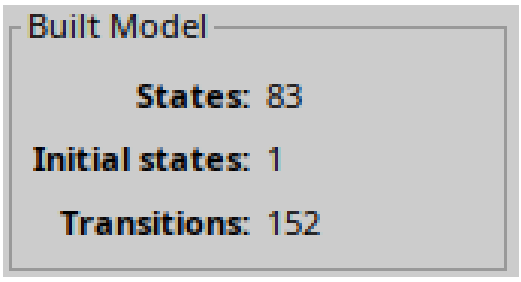


Figure 1: PRISM output showing the number of states in the dtmc model.

e) Use PRISM to verify that the following PCTL formula, which aims at expressing that started jobs are almost surely completed in the future:

$$(task_1 > 0) \Rightarrow \mathbb{P}_{\geq 1}(F(task_1 = 0))$$

See Figure 2.

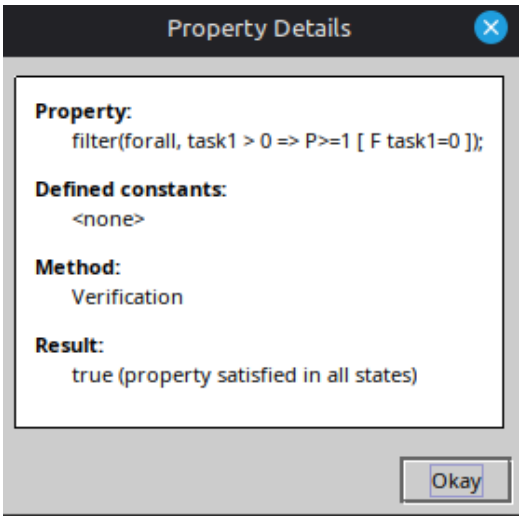


Figure 2: PRISM verification of the PCTL property for eventual job completion.

A03P.2 - Loke S. G. Jørgensen This question is about using PRISM to compute numerical properties of your model.

a) Calculate the transient distribution of the model at time $t = 10$. What is the probability that $Client_1$ does not currently have a job at this time?
We can find the probably of $Client_1$ not having a job (i.e. `state1=0`) at $t = 10$ using property verification in PRISM. The probability is 0.2575, see Figure 3.

b) Plot a graph of the probability that $Client_1$ does not have a job, against time t, for $0 \leq t \leq 10$. Can you given an intuitive explanation of your graph?
See Figure 4. At $t = 0$, the probability is 1, as `state1=0` in the initial state. At $t = 1$, a job will have been created for one of the clients, and since we are using a uniform distribution, the probability that $Client_1$ will be chosen is 0.5. At $t = 2$, the only enabled action is to shift the job created at $t = 1$ to the head of the queue,

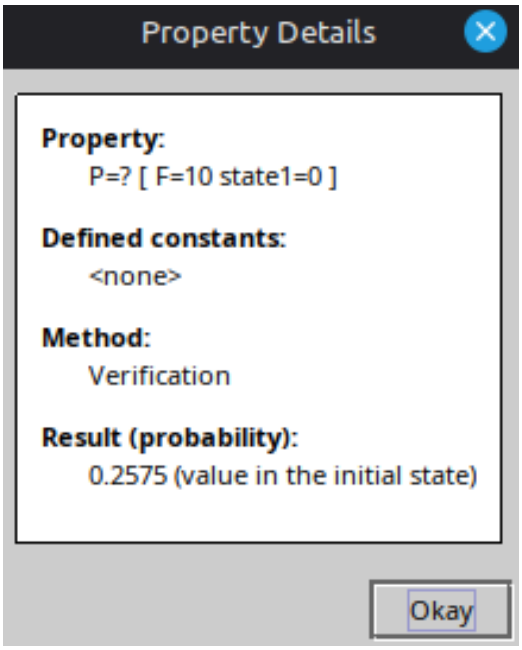


Figure 3: PRISM verification of the PCTL property for eventual job completion.

so the probability is unchanged. The probability then decreases until $t = 5$, when it starts increasing. This is because $t = 4$ is the earliest time a job that was created at $t = 1$ could be finished, in which case the client would again have no job at $t = 5$.

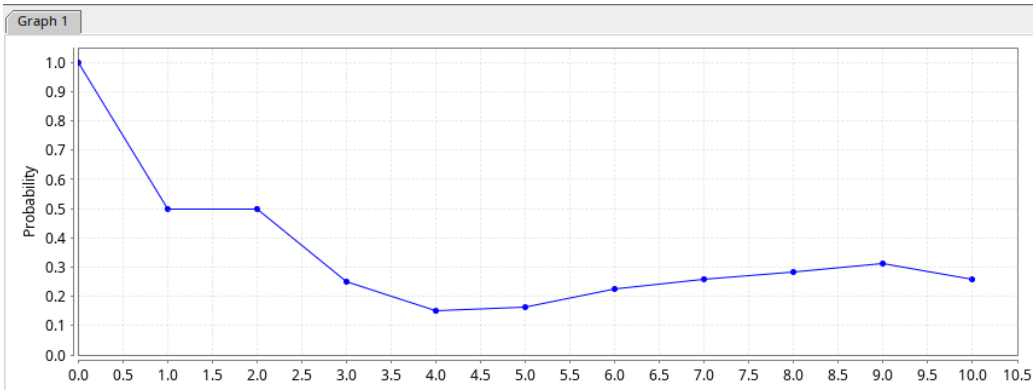


Figure 4: Probability that Client 1 has no job over time.

c) What is the probability that there are no jobs in the queue of the scheduler in the steady state distribution?

We use PRISM's steady-state operator to construct a property that we then verify via the GUI. The probability is 0.016, rounding to three decimals. See Figure 5.

HINT: PRISM offers several functionalities to solve the above tasks. Some of them can be solved through GUI/command-line or using the PRISM Specification Language (see e.g. `https://www.prismmodelchecker.org/manual/PropertySpecification/Introduction`).

A03P.3 This question is about PCTL model checking in PRISM. For each of the following, write down a PCTL formula that captures the query, and use PRISM to determine

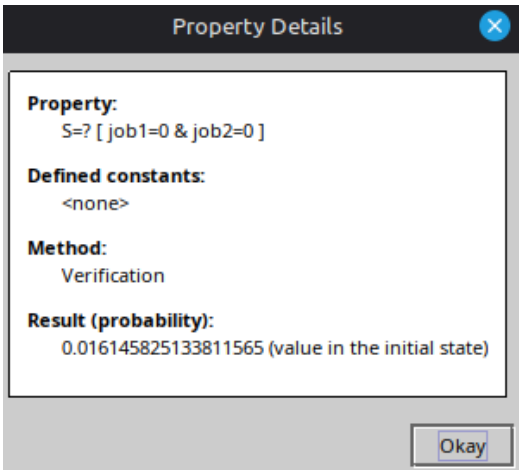


Figure 5: PRISM verification of the steady-state probability that the scheduler queue is empty.

whether the *initial state* of the model satisfies it (provide screenshots):

a) Is the probability greater than 0.2 that $Client_1$ will have have an active job of length greater than 2 in the next time unit?

Provide your answer here. Leave the special color (blue). Figures, tables, code snippets can be placed somewhere else but they need to be referred here.

UPLOAD REQUIRED: a prism property file `A03P2.3.a.props`.

b) Is the probability less than 0.5 that $Client_2$ will create a job of length 5 within 10 time units?

Provide your answer here. Leave the special color (blue). Figures, tables, code snippets can be placed somewhere else but they need to be referred here.

UPLOAD REQUIRED: a prism property file `A03P2.3.b.props`.

c) Is the probability greater than zero that $Client_1$ will always have an active job?

Provide your answer here. Leave the special color (blue). Figures, tables, code snippets can be placed somewhere else but they need to be referred here.

UPLOAD REQUIRED: a prism property file `A03P2.3.c.props`.

d) For each of the above properties, use the '`P=?`' notation in PRISM, to calculate the actual probability of the path formula holding.

Provide your answer here. Leave the special color (blue). Figures, tables, code snippets can be placed somewhere else but they need to be referred here.

UPLOAD REQUIRED: a prism property file `A03P2.3.d.props`.

A03T: Theoretical Problems

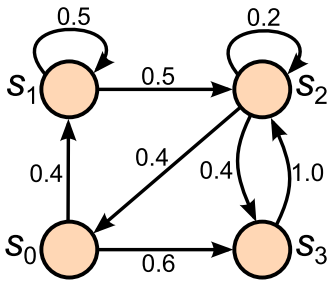


Figure 6: A DTMC

A03T.1 - Maria Mitsi Consider the DTMC in Figure 6, which is shown graphically. The initial state is s_0.

a) Write down the probability transition matrix corresponding to the DTMC, and its initial distribution as a row vector. The state s_i should correspond to the index $i+1$.

The transition matrix is

$$P = \begin{pmatrix} 0 & 0.4 & 0 & 0.6 \\ 0 & 0.5 & 0.5 & 0 \\ 0.4 & 0 & 0.2 & 0.4 \\ 0 & 0 & 1.0 & 0 \end{pmatrix}.$$

Since the initial state is s_0, the initial distribution is

$$\Theta_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix}.$$

b) Compute the transient distribution Θ_3 of the DTMC after 3 time steps. *Hint: you may want to start by computing* Θ_1 *and then* Θ_2.

The transient distribution is computed using

$$\Theta_{n+1} = \Theta_n P.$$

First,

$$\Theta_1 = \Theta_0 P = \begin{pmatrix} 0 & 0.4 & 0 & 0.6 \end{pmatrix}.$$

Next,

$$\Theta_2 = \Theta_1 P = \begin{pmatrix} 0 & 0.2 & 0.8 & 0 \end{pmatrix}.$$

Therefore,

$$\boxed{\Theta_3 = \Theta_2 P = \begin{pmatrix} 0.32 & 0.10 & 0.26 & 0.32 \end{pmatrix}}$$

c) Write down the matrix equation whose fixed point is the steady state solution of the DTMC. Solve this equation system with your favourite method/tool (excluding PRISM).

The steady-state distribution Θ satisfies

$$\Theta P = \Theta, \qquad \sum_{i=0}^{3} \theta_i = 1.$$

Writing $\Theta = (\theta_0, \theta_1, \theta_2, \theta_3)$ gives

$$\begin{cases} \theta_0 = 0.4\theta_2, \\ \theta_1 = 0.4\theta_0 + 0.5\theta_1, \\ \theta_2 = 0.5\theta_1 + 0.2\theta_2 + \theta_3, \\ \theta_3 = 0.6\theta_0 + 0.4\theta_2, \\ \theta_0 + \theta_1 + \theta_2 + \theta_3 = 1. \end{cases}$$

Solving this system gives the steady-state distribution

$$\boxed{\Theta = \begin{pmatrix} \frac{10}{59} & \frac{8}{59} & \frac{25}{59} & \frac{16}{59} \end{pmatrix}}$$

Approximately,

$$\Theta \approx \begin{pmatrix} 0.1695 & 0.1356 & 0.4237 & 0.2712 \end{pmatrix}.$$

A03T.2 - Stefano Giroletti Encode the DTMC in Figure 6 as a PRISM module, using a variable s to represent the state, such that $0 \leq s \leq 3$. Use PRISM to compute the steady state distribution, and check that it agrees with your answer to question 1(c).

The DTMC is encoded in PRISM as follows:

```
dtmc

module client1
  s : [0..3] init 0;

  [] s=0 -> 0.4:(s'=1)
          + 0.6:(s'=3);

  [] s=1 -> 0.5:(s'=1)
          + 0.5:(s'=2);

  [] s=2 -> 0.2:(s'=2)
          + 0.4:(s'=3)
          + 0.4:(s'=0);

  [] s=3 -> 1:(s'=2);

endmodule
```

The steady state distribution computed by PRISM is shown in Figure 7:

$$\Theta = (\theta_0, \theta_1, \theta_2, \theta_3) \approx (0.17,\ 0.14,\ 0.42,\ 0.27),$$

which agrees with the answer to question 1(c).

```
Exporting steady-state probabilities in plain text format below:
0: (0)=0.1694916167456058
1: (1)=0.1355931954330166
2: (2)=0.42372867450100926
3: (3)=0.2711865133203683
```

Figure 7: Steady state distribution computed with PRISM

A03T.3 - Stefano Giroletti For each state s_i, calculate the probability probability of reaching s_2 while avoiding s_3. Explain your solution in terms of the formal semantics based on cylinder sets and double-check your solution against PRISM.

We label the states with $AP = \{end,\ death\}$ and

$$L(s_2) = \{end\}, \qquad L(s_3) = \{death\}, \qquad L(s_0) = L(s_1) = \emptyset.$$

The property to check is $\phi = \neg death\ \mathcal{U}\ end$. For an infinite path π, writing $\pi[k]$ for its k-th state, the semantics is

$$\pi \vDash \phi \iff \exists k \in \mathbb{N}.\ \big(\pi[k] \vDash end\ \wedge\ \forall j < k.\ \pi[j] \vDash \neg death\big).$$

Let $Sat(\phi) = \{\pi \mid \pi \vDash \phi\}$. For each state s_i we want $Pr_{s_i}(\phi) = Pr_{s_i}(Sat(\phi))$, where Pr_{s_i} is the measure with $\iota(s_i) = 1$ and

$$Pr_{s_i}\big(Cyl(t_0 t_1 \dots t_n)\big) = \iota(t_0) \cdot \prod_{k=0}^{n-1} P(t_k, t_{k+1}).$$

For every state we proceed in the same way: first we describe $Paths(s_i)$ with an exhaustive family of cylinder sets, then we check through the semantics which of them satisfy ϕ, then we compute their probabilities, and finally we combine them into the result.

State s_2. Cylinder sets. Every path from s_2 starts with s_2, so $Paths(s_2) = Cyl(s_2)$. Semantics. Taking $k = 0$ we have $\pi[0] = s_2 \vDash end$, and there is no $j < 0$ to check, so the condition on $\neg death$ holds. Hence $Cyl(s_2) \subseteq Sat(\phi)$. Probability. We get

$$Pr_{s_2}(Cyl(s_2)) = \iota(s_2) = 1.$$

Result. Therefore $Pr_{s_2}(\phi) = 1$.

State s_1. Cylinder sets. The only successors of s_1 are s_1 and s_2, so a path either stays in s_1 forever or leaves it towards s_2 after $n-1$ self-loops. This gives the disjoint and exhaustive decomposition

$$Paths(s_1) = \{s_1^\omega\} \ \uplus \biguplus_{n \geq 1} Cyl(s_1^n s_2).$$

Semantics. For a path in $Cyl(s_1^n s_2)$ we take $k = n$: then $\pi[n] = s_2 \vDash end$ and $\pi[j] = s_1 \vDash \neg death$ for all $j < n$, so $Cyl(s_1^n s_2) \subseteq Sat(\phi)$. The path s_1^ω never visits a state satisfying end, so $s_1^\omega \notin Sat(\phi)$. Probabilities. The cylinder set $Cyl(s_1^n s_2)$ requires $n-1$ self-loops, each with probability $\frac{1}{2}$, followed by the move to s_2, again with probability $\frac{1}{2}$:

$$Pr_{s_1}\big(Cyl(s_1^n s_2)\big) = \iota(s_1) \cdot \left(\tfrac{1}{2}\right)^{n-1} \cdot \tfrac{1}{2} = 1 \cdot \left(\tfrac{1}{2}\right)^{n-1} \cdot \tfrac{1}{2} = \left(\tfrac{1}{2}\right)^{n}.$$

The path that stays in s_1 forever has probability bounded by that of staying n steps, for every n:

$$Pr_{s_1}(\{s_1^\omega\}) = \lim_{n\to\infty} \iota(s_1) \cdot \left(\tfrac{1}{2}\right)^n = \lim_{n\to\infty} 1 \cdot \left(\tfrac{1}{2}\right)^n = 0.$$

Result. Only the cylinder sets ending in s_2 satisfy ϕ, and the remaining path has measure 0, so

$$Pr_{s_1}(\phi) = \sum_{n\geq 1} \left(\tfrac{1}{2}\right)^n = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1.$$

State s_3. Cylinder sets. Since $P(s_3, s_2) = 1$, every path from s_3 starts with $s_3 s_2$, hence $Paths(s_3) = Cyl(s_3 s_2)$. Semantics. Here $\pi[0] = s_3 \vDash death$ and $s_3 \nvDash end$. The witness $k = 0$ fails because $\pi[0] \nvDash end$, and any witness $k \geq 1$ fails because it would require $\pi[0] \vDash \neg death$. Hence $Cyl(s_3 s_2) \cap Sat(\phi) = \emptyset$. Probability. We have

$$Pr_{s_3}(Cyl(s_3 s_2)) = \iota(s_3) \cdot P(s_3, s_2) = 1 \cdot 1 = 1,$$

Result. Therefore $Pr_{s_3}(\phi) = Pr_{s_3}(\emptyset) = 0$.

State s_0. Cylinder sets. The successors of s_0 are s_3 and s_1. Decomposing $Cyl(s_0 s_1)$ as done for s_1, we get the disjoint and exhaustive family

$$Paths(s_0) = Cyl(s_0 s_3) \;\uplus\; \{s_0 s_1^\omega\} \;\uplus\; \biguplus_{n\geq 1} Cyl(s_0 s_1^n s_2).$$

Semantics. In $Cyl(s_0 s_3)$ we have $\pi[0] = s_0 \nvDash end$, so $k = 0$ fails; also $\pi[1] = s_3 \nvDash end$, so $k = 1$ fails, and any $k \geq 2$ fails because $\pi[1] = s_3 \vDash death$. Hence $Cyl(s_0 s_3) \cap Sat(\phi) = \emptyset$. In $Cyl(s_0 s_1^n s_2)$ we take $k = n + 1$: then $\pi[n+1] = s_2 \vDash end$ and $\pi[j] \in \{s_0, s_1\} \vDash \neg death$ for all $j \leq n$, so $Cyl(s_0 s_1^n s_2) \subseteq Sat(\phi)$. Finally $s_0 s_1^\omega$ never reaches s_2, so $s_0 s_1^\omega \notin Sat(\phi)$. Probabilities. We compute

$$\begin{aligned}
Pr_{s_0}(Cyl(s_0 s_3)) &= \iota(s_0) \cdot P(s_0, s_3) = 1 \cdot 0.6 = 0.6, \\
Pr_{s_0}(Cyl(s_0 s_1^n s_2)) &= 1 \cdot 0.4 \cdot \left(\tfrac{1}{2}\right)^{n-1} \cdot \tfrac{1}{2} = 0.4 \cdot \left(\tfrac{1}{2}\right)^n, \\
Pr_{s_0}(\{s_0 s_1^\omega\}) &= \lim_{n\to\infty} 1 \cdot 0.4 \cdot \left(\tfrac{1}{2}\right)^{n-1} = 0.
\end{aligned}$$

Result. Only the cylinder sets $Cyl(s_0 s_1^n s_2)$ satisfy ϕ, so

$$Pr_{s_0}(\phi) = \sum_{n\geq 1} 0.4 \cdot \left(\tfrac{1}{2}\right)^n = 0.4 \cdot \frac{\frac{1}{2}}{1-\frac{1}{2}} = 0.4.$$

Summary,

$$Pr_{s_0}(\phi) = 0.4, \qquad Pr_{s_1}(\phi) = 1, \qquad Pr_{s_2}(\phi) = 1, \qquad Pr_{s_3}(\phi) = 0.$$